

CASE STUDY

Google Uses SiFive Intelligence X280 IP To Drive Its Datacenter TPU AI Acceleration Engine

THE CHALLENGE

Google, a global leader in AI innovation, sought to enhance their Tensor Processing Units (TPUs) for data center AI workloads, including image classification, recommendation systems, and large-scale machine learning tasks. They needed a flexible, high-performance AI compute host to manage their custom MXU (Matrix Multiply Unit) accelerators while maintaining efficiency, programmability, and low-latency communication. Traditional methods like PCIe introduced high latency (hundreds to thousands of cycles), for complex compute tasks and maintenance code not suited for accelerators. Google required a compact, scalable solution that could seamlessly integrate with their MXU architecture to enable diverse AI tasks acceleration without compromising performance or power.

THE SOLUTION

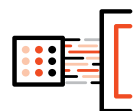
At the AI Hardware Summit in Santa Clara, Google described why, after rigorous benchmarking, they selected SiFive Intelligence X280 RISC-V-based processor IP as the Accelerator Control Unit (ACU) for their TPU. The X280, paired with SiFive's VCIX bus, delivers efficient command and control, offloading complex AI tasks like image classification to Google's MXU with low-latency, high-bandwidth communication. This ACU approach optimizes TPU performance for real-time machine learning workloads in data centers. The VCIX bus ensures seamless coordination between the X280 and MXU, enhancing computational efficiency. It enables Google to focus on their strengths in MXU design, delivering scalable, programmable solutions that position their TPUs for leadership in AI innovation.



Efficient 64-bit
Scalar Compute



Wide 512-bit
Vector Compute



VCIX Connection
to External MXU



WHY SIFIVE

Google's adoption of SiFive Intelligence X280 followed an extensive evaluation of IP options for their TPU architecture. The X280's compact, high-performance vector processing, paired with VCIX's 10-cycle latency, proved ideal for managing AI workloads like image classification and machine learning. Its full Linux support with virtual memory and cache coherence met Google's need for a programmable compute host, enabling seamless integration with their MXU accelerators and ensuring a competitive edge in data center AI.

[Read Blog](#)

CUSTOMER RESULTS



High-Performance Vector Processing

32, 512-bit vector registers with up to 4096-bit operations accelerate image classification and machine learning in data centers



Flexible Accelerator Integration

VCIX enables 10-cycle, high-bandwidth coupling with Google's MXU, boosting TPU efficiency



Flexibility and Programmability

Supports complete software stacks with hypervisor and virtual memory for diverse AI workloads



Compact and Efficient

Optimized design minimizes power, enabling scalable, cost-effective TPU solutions

MORE INFORMATION

Download the Intelligence X280 Product Brief to find out more about X280, read the blog to find out more about Google's TPU use case or connect to sales to ask us questions about how X280 or other SiFive Intelligence products could solve your unique challenges and needs.

[X280 Product Brief](#)[Contact Sales](#)

ABOUT SIFIVE

SiFive leads the RISC-V revolution, delivering cutting-edge processor IP that powers the future of edge AI and data center innovation. With solutions like SiFive Intelligence X280, we enable customers like Google to create high-performance, power-efficient SoCs for AI workloads, including image classification and machine learning. Founded to drive innovation through open-standard RISC-V architecture, SiFive provides scalable, customizable IP that accelerates development of advanced AI solutions. Our expertise empowers global leaders to build compact, cost-effective systems that redefine industries.

©2025 SiFive, Inc. All rights reserved.

SiFive and the SiFive logo are trademarks or registered trademarks of SiFive, Inc. Certain products or brand names that are not SiFive's could be trademarks or registered trademarks of their respective owners.